

Support-Reduced Markov Update from the Two-Leg Contraction

Purpose

This note records the explicit reduction used by the production `run_markov_circuit(...)` code when the Wannier modes have finite support (`nshell` finite). The point is that one does not need to build a full-system Choi covariance for each local channel. Instead, the two-leg Gaussian contraction can be reduced analytically to a solve on the support block of the measured mode.

1. Generic two-leg composition formula

Suppose

$$\Psi^{(1)} = \begin{pmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{pmatrix}, \quad \Psi^{(2)} = \begin{pmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{pmatrix}. \quad (1)$$

Contracting the output leg of $\Psi^{(1)}$ with the input leg of $\Psi^{(2)}$ gives

$$\Psi^{(2)} \circ \Psi^{(1)} = \begin{pmatrix} A_{11} & 0 \\ 0 & B_{22} \end{pmatrix} - \begin{pmatrix} A_{12} & 0 \\ 0 & B_{21} \end{pmatrix} \begin{pmatrix} A_{22} & -\mathbb{I} \\ -\mathbb{I} & B_{11} \end{pmatrix}^{-1} \begin{pmatrix} A_{21} & 0 \\ 0 & B_{12} \end{pmatrix}. \quad (2)$$

Equivalently,

$$\Psi^{(2)} \circ \Psi^{(1)} = M - NK^{-1}N^\dagger, \quad (3)$$

with

$$M = \text{diag}(A_{11}, B_{22}), \quad (4)$$

$$N = \text{diag}(A_{12}, B_{21}), \quad (5)$$

$$K = \begin{pmatrix} A_{22} & -\mathbb{I} \\ -\mathbb{I} & B_{11} \end{pmatrix}. \quad (6)$$

For Hermitian Gaussian covariances, $A_{21} = A_{12}^\dagger$ and $B_{12} = B_{21}^\dagger$, so the rightmost block factor is exactly $N^\dagger$.

2. Dense state update from a two-leg Gaussian tensor

Let the top-layer covariance be

$$G \in \mathbb{C}^{N \times N}. \quad (7)$$

If a two-leg Gaussian tensor has block form

$$\Psi = \begin{pmatrix} \Psi_{11} & \Psi_{12} \\ \Psi_{21} & \Psi_{22} \end{pmatrix}, \quad (8)$$

then contraction with the state tensor gives the dense update

$$G' = \Psi_{22} - \Psi_{21}(G\Psi_{11} - \mathbb{I})^{-1}G\Psi_{12}. \quad (9)$$

For a rank-one projector

$$P = |\chi\rangle\langle\chi|, \quad Q = \mathbb{I} - P, \quad (10)$$

the measurement branches are represented by

$$\Psi_{\text{occ}} = \begin{pmatrix} -P & Q \\ Q & P \end{pmatrix}, \quad \Psi_{\text{emp}} = \begin{pmatrix} P & Q \\ Q & -P \end{pmatrix}. \quad (11)$$

This is exactly the formula above with the first tensor taken to be the state tensor

$$\Sigma(G) = \begin{pmatrix} 0 & \mathbb{I} \\ \mathbb{I} & G \end{pmatrix}, \quad (12)$$

so that $A_{11} = 0$, $A_{12} = A_{21} = \mathbb{I}$, $A_{22} = G$, and the lower-right block of (??) becomes (??). Substituting the measurement tensors into (??) yields the exact dense measurement update.

3. Support decomposition

Assume the mode χ has support only on a subset s of indices. Let r be the complement. After reordering indices so that s comes first, write

$$G = \begin{pmatrix} G_{ss} & G_{sr} \\ G_{rs} & G_{rr} \end{pmatrix}, \quad P = \begin{pmatrix} p & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} q & 0 \\ 0 & \mathbb{I}_r \end{pmatrix}, \quad (13)$$

where

$$p = \chi_s \chi_s^\dagger, \quad q = \mathbb{I}_s - p. \quad (14)$$

The production code stores exactly this data: the support indices, the complement indices, and the restricted mode vector χ_s .

4. Occupied branch

For the occupied branch, set

$$H_{11} = -P, \quad H_{21} = Q, \quad H_{22} = P. \quad (15)$$

Then (??) becomes

$$G' = P - QX, \quad X = (GH_{11} - \mathbb{I})^{-1}GH_{21}^\dagger. \quad (16)$$

In block form,

$$A_+ := GH_{11} - \mathbb{I} = \begin{pmatrix} -(\mathbb{I}_s + G_{ss}p) & 0 \\ -G_{rs}p & -\mathbb{I}_r \end{pmatrix}, \quad B := GH_{21}^\dagger = GQ = \begin{pmatrix} G_{ss}q & G_{sr} \\ G_{rs}q & G_{rr} \end{pmatrix}. \quad (17)$$

Define

$$K_+ := \mathbb{I}_s + G_{ss}p. \quad (18)$$

Writing

$$X = \begin{pmatrix} X_{11} & X_{12} \\ X_{21} & X_{22} \end{pmatrix}, \quad Y := -X_{12}, \quad Z := -X_{11}, \quad (19)$$

the first block row gives the two support solves

$$K_+Y = G_{sr}, \quad K_+Z = G_{ss}q. \quad (20)$$

The updated blocks are then

$$G'_{ss} = p + qZ, \quad (21)$$

$$G'_{sr} = qY, \quad (22)$$

$$G'_{rr} = G_{rr} - G_{rs}pY, \quad (23)$$

with $G'_{rs} = (G'_{sr})^\dagger$.

So the only inverse information needed is the solution of (??) on the support block. No inverse of the projector p or q is required.

5. Empty branch

For the empty branch, set

$$H_{11} = P, \quad H_{21} = Q, \quad H_{22} = -P. \quad (24)$$

Now

$$A_- := GH_{11} - \mathbb{I} = \begin{pmatrix} -(\mathbb{I}_s - G_{ss}p) & 0 \\ G_{rs}p & -\mathbb{I}_r \end{pmatrix}, \quad (25)$$

so define

$$K_- := \mathbb{I}_s - G_{ss}p. \quad (26)$$

With the same definitions of Y and Z , the support solves are

$$K_-Y = G_{sr}, \quad K_-Z = G_{ss}q. \quad (27)$$

The updated blocks are

$$G'_{ss} = -p + qZ, \quad (28)$$

$$G'_{sr} = qY, \quad (29)$$

$$G'_{rr} = G_{rr} + G_{rs}pY, \quad (30)$$

again with $G'_{rs} = (G'_{sr})^\dagger$.

6. Reset / pump channel

The deterministic reset channel has the affine form

$$G' = QGQ + sP, \quad s \in \{+1, -1\}. \quad (31)$$

In support blocks this is immediate:

$$G'_{ss} = qG_{ss}q + sp, \quad (32)$$

$$G'_{sr} = qG_{sr}, \quad (33)$$

$$G'_{rr} = G_{rr}, \quad (34)$$

with $G'_{rs} = (G'_{sr})^\dagger$, $H_{22} =$

7. What the production code actually does

For finite `nshell`, the production code does not construct a local Choi covariance and then re-derive the Schur complement numerically at each site. Instead, it implements the reduced formulas above directly:

- solve on the support with K_+ or K_- ,
- update the support-support and support-rest blocks,
- add the induced correction to the rest-rest block,
- use the affine support formula directly for the reset step.

This is mathematically equivalent to the dense two-leg contraction specialized to a compactly supported rank-one projector channel.

8. Numerical verification

A local numerical check was run in this repository on a small `classA.U1FGTN(4,4,DW=True,nshell=1)` instance. For one site and twenty random Hermitian test matrices G , the reduced formulas were compared against the dense full-space update:

- measurement branch maximum entrywise discrepancy: 1.8×10^{-12} ,
- reset branch maximum entrywise discrepancy: 4.5×10^{-16} .

These are consistent with roundoff-level agreement.